

Rising Waters: A Machine Learning Approach to Flood

PredictionProject Description:

Floods are among the most devastating natural disasters, causing loss of lives, property damage, and environmental destruction every year. Accurate flood prediction plays a vital role in disaster preparedness and emergency response. Traditional forecasting methods often fail to provide timely and precise predictions due to the complexity of weather patterns.

This project presents an intelligent flood prediction system using Machine Learning algorithms such as Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost. Historical weather data containing annual rainfall, seasonal rainfall, cloud cover, humidity, and other meteorological parameters is used to train the models.

After comparing all models, XGBoost achieved the highest accuracy of **96.55%** and was selected as the final prediction model. The trained model is integrated with a Flask web application where users can enter weather parameters and instantly obtain flood prediction results.

The application is designed for future deployment on IBM Cloud, enabling disaster management authorities to monitor flood risks from anywhere.

Scenarios

Scenario 1 – Early Flood Warning

A weather officer enters current rainfall, humidity, river water level, and cloud cover values into the system. The model predicts a high probability of flooding, allowing authorities to issue early evacuation alerts.

Scenario 2 – Disaster Management

During heavy monsoon season, disaster management officials use the application to analyze multiple districts. The system predicts flood-prone regions, helping authorities allocate rescue teams and emergency supplies efficiently.

Scenario 3 – Model Evaluation

Government analysts evaluate the trained model using historical flood datasets. Among all classification algorithms, XGBoost achieves **96.55% accuracy**, proving its effectiveness for operational flood prediction.

Technical Architecture

Description

The Flood Prediction System follows a Machine Learning-based three-layer

architecture.

Presentation Layer

- HTML
- CSS
- JavaScript
- Flask Templates

Application Layer

- Flask Framework
- Prediction Engine
- Machine Learning Models

Data Layer

- Flood Dataset (CSV)
- Trained Model (.pkl)
- Weather Data

The user enters weather parameters through the web interface. Flask processes the request, sends the data to the trained XGBoost model, and displays the prediction result.

Pre-Requisites

- Python Programming
- Machine Learning Basics
- Flask Framework
- NumPy
- Pandas
- Scikit-learn
- Matplotlib
- XGBoost
- HTML
- CSS
- IBM Cloud Basics

Project Workflow

Milestone 1

Dataset Collection

- Download flood dataset from Kaggle.
- Understand dataset features.
- Prepare development environment.

Milestone 2

Data Analysis

- Perform Exploratory Data Analysis.
- Visualize rainfall patterns.
- Identify missing values and outliers.

Milestone 3

Data Preprocessing

- Handle missing values.
- Feature Scaling.
- Split Training and Testing datasets.

Milestone 4

Model Development

- Train Decision Tree.
- Train Random Forest.
- Train KNN.
- Train XGBoost.
- Compare accuracy.

Milestone 5

Flask Application

- Create Home Page.
- Prediction Form.
- Display Flood Prediction.
- Display No Flood Prediction.

Milestone 6

Deployment

- Save trained model.
- Deploy application on IBM Cloud.
- Perform testing.

Conclusion

The Flood Prediction System successfully predicts flood risk using Machine Learning algorithms. Among all models, XGBoost achieved the highest accuracy of **96.55%**. The Flask web application provides a simple interface for entering weather parameters and obtaining prediction results instantly. This project demonstrates how Artificial Intelligence can support disaster management by providing accurate and timely flood predictions.